

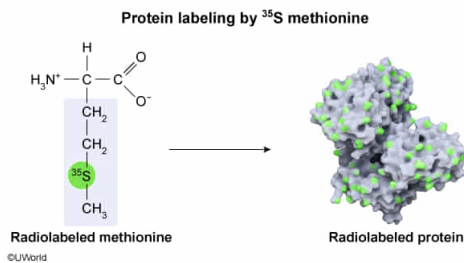
Researchers quantified the amount of protein produced by cultured cells during a one-hour period by detecting the radioisotope ^{35}S . Prior to this period, the cells were most likely provided with radiolabeled:

- ☐ A. asparagine. [2%]
☒ B. methionine. [77%]
☐ C. threonine. [3%]
☐ D. serine. [15%]

Correct

77% answered correctly

Explanation:



Cells continuously produce proteins by combining free amino acids into polypeptide chains. Once they are produced, proteins typically remain intact for hours to days before being degraded. Researchers interested in comparing the amounts of protein produced by cells in a given period must disregard any proteins that were already present before the assay began.

One way to **quantify the amount of protein** produced during a specified period is to provide the cells with **amino acids** that **contain a radioactive atom** for the duration of the assay. Proteins that are synthesized during the specified time frame will incorporate the radiolabeled amino acids into their **primary structure**. These proteins can then be detected by measuring radioactive emissions, whereas proteins that were synthesized prior to the period of interest will contain no radioisotopes and will not be detected.

Of the 20 standard amino acids, only **cysteine** and **methionine** contain sulfur atoms. Either of these amino acids could be labeled with ^{35}S to facilitate detection. However, methionine is the only amino acid among the given choices that contains sulfur, so the cells were most likely **provided with radiolabeled methionine**.

(Choices A, C, and D) Asparagine, threonine, and serine contain no sulfur atoms, so they cannot contain the radioisotope ^{35}S .

Educational objective:

Protein synthesis may be quantified by detection of radiolabeled amino acids. ^{35}S is a radioactive isotope that is commonly used to label proteins, typically as a component of the amino acid methionine.

Time spent:0 sec

Subject:
Biochemistry

QID:402696

System:
Amino Acids and Proteins